Problem Sheet: Week 10 - solutions

1. (a) The posterior distribution is given by (seen in class)

$$\pi(p|x) \propto p^x(1-p)^{n-x}p^{\alpha-1}(1-p)^{\beta-1} = p^{x+\alpha-1}(1-p)^{n-x+\beta-1}$$

Therefore the posterior distribution is Beta$(x+\alpha, n+\beta-x)$, and the posterior mean is

$$\mathbf{E}\,(p|x) = \frac{x+\alpha}{\alpha+\beta+n} = Z\cdot\frac{x}{n} + (1-Z)\cdot\frac{\alpha}{\alpha+\beta},$$

where $Z = n/(\alpha+\beta+n)$.

(b) We know that

$$\frac{\alpha}{\alpha+\beta} = 0.005; \quad \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = 0.0015^2.$$

By solving the system of equations we find $\alpha = 11.0506$ and $\beta = 2199.06$. Thus, the credibility estimate is $Z = 10000/12210.1106 = 0.819$.

The credibility estimate for p is therefore

$$0.819 \times 0.0056 + 0.181 \times 0.005 = 0.0055.$$

2. (a) The posterior distribution is (seen in class)

$$\pi(\theta|\underline{\mathrm{x}}) \propto \exp\left\{-\frac{n(\bar{x}-\theta)^2}{2\sigma_1^2}\right\}\exp\left\{-\frac{(\theta-\mu)^2}{2\sigma_2^2}\right\}$$

where $\bar{x} = \sum x_i/n$.

By expanding the expressions and simplifying, we find that

$$\begin{aligned}\pi(\theta|\underline{\mathrm{x}}) &\propto \exp\left\{\frac{n\bar{x}}{\sigma_1^2}\theta - \frac{n}{2\sigma_1^2}\theta^2 - \frac{1}{2\sigma_2^2}\theta^2 + \frac{\mu}{\sigma_2^2}\theta\right\}\\ &= \exp\left\{-\frac{n\sigma_2^2+\sigma_1^2}{2\sigma_1^2\sigma_2^2}\theta^2 + \frac{n\bar{x}\sigma_2^2+\mu\sigma_1^2}{\sigma_1^2\sigma_2^2}\theta\right\}\\ &= \exp\left\{A\theta^2 + B\theta\right\}\end{aligned}$$

where

$$A = -\frac{n\sigma_2^2+\sigma_1^2}{2\sigma_1^2\sigma_2^2} \quad , \quad B = \frac{n\bar{x}\sigma_2^2+\mu\sigma_1^2}{\sigma_1^2\sigma_2^2}.$$

Now, complete the square and ignore terms not involving θ. The expression is found to be proportional to

$$\pi(\theta|\underline{\mathrm{x}}) \propto \exp\left\{A\left(\theta + \frac{B}{2A}\right)^2\right\}$$
$$= \exp\left\{-\frac{n\sigma_2^2+\sigma_1^2}{2\sigma_1^2\sigma_2^2}\left[\theta - \left(\frac{\mu\sigma_1^2+n\bar{x}\sigma_2^2}{\sigma_1^2+n\sigma_2^2}\right)\right]^2\right\}$$

Therefore

$$\theta|\underline{\mathrm{x}} \sim \mathrm{N}\left(\frac{\mu\sigma_1^2+n\bar{x}\sigma_2^2}{\sigma_1^2+n\sigma_2^2}, \frac{\sigma_1^2\sigma_2^2}{n\sigma_2^2+\sigma_1^2}\right)$$

and the posterior mean is

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{\mu\sigma_1^2+n\bar{x}\sigma_2^2}{\sigma_1^2+n\sigma_2^2} = Z\bar{x} + (1-Z)\mu$$

where $Z = n/(n+\sigma_1^2/\sigma_2^2)$.

(b) ➢ The credibility factor is an increasing function of n, i.e. the more data we have the more weight is given to data from the risk.

➢ The credibility factor is a decreasing function of σ_1^2, i.e. within risk variance. The more uncertain we are in data from the risk, the less weight is given to the risk.

➢ The credibility factor is an increasing function of σ_2^2, i.e. between risk variance. The more uncertain we are about the collateral risk (similar but not identical risk), the more weight is given to data from the risk.

3. (a) First note that the prior on θ is improper. Let us choose $\pi(\theta) = C$ for some $C > 0$. By mimicking Bayes theorem, we define the posterior distribution of θ as

$$\pi(\theta|\underline{\mathrm{x}}) = \frac{C\exp\{-n\theta\}\theta^{n\bar{x}}}{\int_0^\infty C\exp\{-n\theta'\}(\theta')^{n\bar{x}}d\theta'} = \frac{\exp\{-n\theta\}\theta^{n\bar{x}}}{\int_0^\infty \exp\{-n\theta'\}(\theta')^{n\bar{x}}d\theta'},$$

noting that the function of data ($\prod x_i!$) cancels in the fraction.

Thus, the posterior mean is simply

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \int_0^\infty \theta \cdot \frac{\exp\{-n\theta\}\theta^{n\bar{x}}}{\int_0^\infty \exp\{-n\theta'\}(\theta')^{n\bar{x}}d\theta'}d\theta = \frac{\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}+1}\,d\theta}{\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}}\,d\theta}$$

We now recognise that both the integrands resemble the form of a gamma distribution, i.e., we know that the normalising constant of the Gamma(α,β) distribution is

$$\int_0^\infty x^{\alpha-1}\exp\{-\beta x\}dx = \frac{\Gamma(\alpha)}{\beta^\alpha}.$$

The integrals can therefore be simplified to

$$\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}+1}\, d\theta = \frac{\Gamma(n\bar{x}+2)}{n^{n\bar{x}+2}}$$

$$\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}}\, d\theta = \frac{\Gamma(n\bar{x}+1)}{n^{n\bar{x}+1}}$$

Hence, the posterior mean can be simplied to

$$\begin{aligned}
\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) &= \frac{\Gamma(n\bar{x}+2)}{n^{n\bar{x}+2}} \cdot \frac{n^{n\bar{x}+1}}{\Gamma(n\bar{x}+1)} \\
&= \frac{\Gamma(n\bar{x}+2)}{n\Gamma(1)\Gamma(n\bar{x}+1)} \qquad \because \Gamma(1) = 1 \\
&= \frac{1}{nB(1, n\bar{x}+1)}
\end{aligned}$$

NOTE: It is not correct to say that $\Gamma(n\bar{x}+2) = (n\bar{x}+1)!$ or $\Gamma(n\bar{x}+1) = (n\bar{x})!$ since $n\bar{x}$ is not necessarily a positive integer.

(b) i. When $\bar{x} = 5$, we have

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{\Gamma(27)}{5 \cdot \Gamma(26)} = 5.2,$$

using the fact that $\Gamma(a) = (a-1)!$ when a is a natural number.

ii. When $\bar{x} = 6.4$, we have

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{\Gamma(34)}{5 \cdot \Gamma(33)} = 6.6.$$

iii. When $\bar{x} = 8.7$, we can use the `beta()` function in R to compute the following

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{1}{5 \cdot B(1, 44.5)} = 8.9.$$

(c) We first evaluate the beta function as follows:

$$B(1, n\bar{x}+1) = \int_0^1 (1-t)^{n\bar{x}} dt = \left[-\frac{(1-t)^{n\bar{x}+1}}{n\bar{x}+1}\right]_0^1 = \frac{1}{n\bar{x}+1}$$

Thus, the posterior mean is

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{1}{nB(1, n\bar{x}+1)} = \frac{n\bar{x}+1}{n} = \bar{x} + \frac{1}{n}$$

Alternative solution: It is possible to apply integration by parts at the outset

$$\begin{aligned}\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}+1}\,d\theta &= \int_0^\infty \left(-\frac{1}{n}\exp\{-n\theta\}\right)' \theta^{n\bar{x}+1}\,d\theta \\ &= \lim_{M\to\infty}\left[-\frac{1}{n}\exp\{-n\theta\}\theta^{n\bar{x}+1}\right]_0^M + \int_0^\infty \frac{n\bar{x}+1}{n}\exp\{-n\theta\}\theta^{n\bar{x}}d\theta \\ &= \frac{n\bar{x}+1}{n}\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}}d\theta\end{aligned}$$

Thus, the posterior mean is

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}+1}\,d\theta}{\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}}\,d\theta} = \frac{n\bar{x}+1}{n} = \bar{x}+\frac{1}{n}$$

4. (a)

$$\mathbf{E}\left[m(\theta)\right] = \frac{87+100+135+112+116+110}{6} = 110$$

(b)

$$\mathbf{E}\left[s^2(\theta)\right] = \frac{692+1361+1015+638+859+841}{6} = 901$$

(c)

$$\mathbf{Var}\left[m(\theta)\right] = \frac{(87-110)^2+(100-110)^2+\cdots+(110-110)^2}{6-1} - \frac{901}{8} = 146.175$$

(d) The credibility factor is calculated as

$$Z = \frac{8}{8+\frac{901}{146.175}} = 0.565.$$

Thus, the credibility estimate of the pure premium is

$$0.565\times 135 + (1-0.565)\times 110 = 124.12$$